MANIKANTA PERUMALLA

manikanta.bhu.stcomp@gmail.com | 9390433680 | [GitHub](#) | [LinkedIn](#)

Professional Summary

Data Science professional with strong foundations in Statistics, Machine Learning, NLP, and Generative AI. Experienced in building analytical and LLM-powered solutions using Python, SQL, and modern GenAI frameworks. Skilled in feature engineering, model development, and transforming structured and unstructured data into actionable insights.

Technical Skills

Programming: Python (Numpy, Pandas, Scikit-learn, Matplotlib, Tensorflow, Pytorch), SQL

Data Engineering: Databricks, Pyspark, Data Pipelines, Azure Data Lakehouse Storage gen2

Machine Learning: Supervised Learning (Regression and Classification), Unsupervised Learning, Ensemble Learning (Bagging Techniques, XG Boost, LGBM) Statistical Analysis.

Data Science: EDA, Data Cleaning, Feature Engineering, Data Validation, Data Modelling, Model Evaluation

Gen AI: LLMs, RAG Pipelines, Prompt Engineering, LangChain, Vector Search, Embeddings

Soft Skills: Problem-Solving, Communication, Time Management, Team Work

Professional Experience

Data Scientist — ValueLabs *Sep 2024 – Present*

- Worked on GenAI, NLP, and machine learning workflows supporting AI-driven analytical solutions.
- Performed data pre-processing, statistical analysis, and feature engineering for model development.
- Assisted in model evaluation, optimization, and deployment-oriented workflows.
- Collaborated with business and technical stakeholders to translate requirements into AI solutions.
- Worked with version control (Git) and structured analytical development workflows.

Subject Matter Expert (Freelancer) — Chegg India *Apr 2022 – May 2023*

- Delivered solutions in statistics, regression, sampling, and machine learning problem-solving.
- Reviewed analytical solutions ensuring statistical correctness and clarity.

Projects

Marketplace Analytics & GenAI Assistant

- Developed a RAG-based LLM application for answering customer queries using internal knowledge bases.
- Built data ingestion and preprocessing pipelines for large-scale datasets.
- Implemented embeddings-based retrieval workflows enabling context-aware responses.
- Applied data validation and feature preparation for analytics and AI workflows.
- Automated reporting and data tracking to improve operational efficiency.

Provider Fraud Detection | US Healthcare Insurance

- Developed an end-to-end fraud detection system targeting fraudulent healthcare providers within US insurance networks, aimed at reducing financial losses and protecting policy integrity.
- Conducted thorough data preparation and feature engineering on insurance claims data to identify patterns and anomalies indicative of provider-level fraudulent behaviour.
- Built and evaluated predictive machine learning models to classify providers as fraudulent or legitimate, enabling proactive risk flagging before claim payouts.
- Structured the project across a complete data science lifecycle — from raw data ingestion and preprocessing through model training, testing, and generating actionable predictions.

Education

M.Sc. Statistics and Computing | Banaras Hindu University | 8.01 *2021 – 2023*

B.Sc. Statistics & Computers | Vikas Degree College | 8.66 *2018 – 2021*

Certification

Certified Data Science Program (NASSCOM Accredited) – 360DigitMG